Development and Validation of a Two-Input Transfer Function Analysis for Cerebral Blood Flow Dynamics

Abstract

Background and Aim: Transfer Function Analysis (TFA) of dynamic cerebral autoregulation (dCA) commonly treats mean arterial pressure (MAP) as the single input affecting cerebral blood flow velocity (CBFV), although end-tidal carbon dioxide ($PETCO_2$) can also influence CBFV. When MAP and $PETCO_2$ fluctuate together, separate single-input-single-output (SISO) estimates may not distinguish their contributions. We aimed to develop and evaluate a two-input TFA framework that jointly estimates MAP-to-CBFV and $PETCO_2$-to-CBFV dynamics.

Methods: The framework applies consistent preprocessing and spectral estimation before solving the MAP and $PETCO_2$ transfer functions together. We will apply SISO and multiple-input-single-output (MISO) analyses to resting baseline recordings from cognitively normal (NC) adults and compare pathway estimates, coherence measures, and numerical diagnostics. Because the true pathways are unknown in physiological recordings, we will also use synthetic time series generated from prespecified transfer functions to evaluate pathway-recovery error across variations in recording duration, noise, timing alignment, input relationships, and estimator settings. Measurable properties of the NC recordings will be compared with the tested simulation conditions.

Results: [To be completed after the final NC analysis and full paper-profile simulation.]

Conclusion: By combining application to physiological recordings with known-truth validation, this study will define the conditions under which two-input TFA provides interpretable MAP- and $PETCO_2$-related estimates of CBFV dynamics.

Introduction

Adequate regulation of cerebral blood flow (CBF) is essential for normal brain function. Among the multiple mechanisms involved in CBF regulation, responses to changes in arterial blood pressure and the partial pressure of arterial carbon dioxide ($PaCO_2$) are particularly important. Cerebral autoregulation (CA) refers to the ability of the cerebral vasculature to buffer the effects of changes in arterial blood pressure on CBF. In response to an increase in blood pressure, cerebrovascular constriction occurs, thereby limiting excessive cerebral perfusion. In contrast, a decrease in blood pressure evokes cerebral vasodilation to protect against cerebral hypoperfusion. Cerebral vasomotor CO_2 reactivity (CVMR) refers to the ability of the cerebral vasculature to adjust vascular tone in response to changes in $PaCO_2$, thereby modulating CBF and contributing to brain tissue pH homeostasis. An increase in $PaCO_2$ lowers pH (acidosis),

leading to cerebral vasodilation and an increase in CBF. The increased CBF facilitates CO_2 washout and limits the decrease in pH. Conversely, a decrease in $PaCO_2$ raises pH (alkalosis), leading to cerebral vasoconstriction and a decrease in CBF. The reduced CBF limits CO_2 washout and attenuates the increase in pH. **Alterations in CA and CVMR have been reported with aging and in various pathophysiological conditions, raising interest in their quantification as potential markers of cerebrovascular dysfunction.**

Cerebral autoregulation can be broadly characterized as static or dynamic. Static CA describes the ability of the cerebral vasculature to maintain relatively stable CBF across sustained changes in steady-state blood pressure. Its assessment generally requires the establishment of different steady-state blood pressure levels, often through pharmacological manipulation interventions that induce relatively large and sustained changes in blood pressure. In contrast, dynamic cerebral autoregulation (dCA) describes the ability of the cerebral vasculature to buffer transient changes in CBF in response to rapid fluctuations in blood pressure that occur over seconds to minutes. Because dCA can be assessed noninvasively from spontaneous blood pressure and CBF fluctuations under resting conditions, it has been widely quantified in both physiological and clinical research. Single-input, Single-output (SISO) Transfer Function analysis (~~The brain requires continuous blood flow even as arterial pressure changes from beat to beat. Cerebral autoregulation describes the physiological processes that limit how strongly these pressure changes alter cerebral perfusion (Regulation of Cerebral Blood Flow in Humans). Dynamic cerebral autoregulation focuses on responses that occur over seconds to minutes.~~ TFA) is widely~~commonly~~ used to quantify dCA based on linear system theory, describing the relationship between ABP (input) and CBF (output) in the frequency domain. Using TFA, dCA is quantified with three metrics: gain, phase, and coherence. ~~study these responses by treating MAP as the input and CBFV as the output (Transfer Function Analysis of Dynamic Cerebral Autoregulation). The resulting transfer function is complex-valued.~~ Gain describes how strongly changes in MAP appear in CBFV (i.e., magnitude relation), while phase describes their timing relationship (i.e., temporal difference) at each frequency. Coherence quantifies the strength of linearity between ABP and CBF fluctuations and ranges from 0 to 1, with a higher coherence value indicating a stronger linear relation and a higher signal-to-noise ratio for estimating TFA gain and phase. Importantly, CBF also responds dynamically to fluctuations in arterial CO_2. **Importantly, CBF also responds dynamically to fluctuations in arterial CO_2. Thus, CBF dynamics may be shaped by concurrent cerebrovascular responses to fluctuations in both blood pressure and CO_2.** ~~Ordinary coherence describes how consistently the two signals are linearly related.~~

Next, raise the potential problem with conventional SISO TFA.

A MAP-only model is useful, but MAP is not the only input that can influence CBFV. Carbon dioxide is a potent vasodilator with a strong effect on cerebrovascular tone, and $PETCO_2$ provides a practical breath-to-breath surrogate for changes in arterial carbon dioxide during resting recordings (Regulation of the Cerebral Circulation by Arterial Carbon Dioxide). In a MAP-only SISO model, $PETCO_2$ is omitted. If MAP and $PETCO_2$ share frequency content, MAP can carry information about $PETCO_2$, and part of the $PETCO_2$-related CBFV response may be attributed to MAP. The same problem applies to a $PETCO_2$-only SISO model. Therefore, two separate SISO models describe pairwise relationships, but they do not distinguish the contribution of one measured input from the other.

Previous studies have shown the potential value of modeling MAP and $PETCO_2$ together. Panerai et al. used multivariate finite impulse response filters to estimate the contributions of beat-to-beat MAP and breath-to-breath $PETCO_2$ fluctuations to resting CBFV variability (Multivariate Dynamic Analysis of Cerebral Blood Flow Regulation). MAP explained a substantial portion of the measured CBFV variability, and adding $PETCO_2$ reduced the model mean squared error. These results suggest that spontaneous $PETCO_2$ fluctuations can help explain resting CBFV variability that is not captured by pressure alone.

However, lower CBFV prediction error does not necessarily mean that the separate MAP and $PETCO_2$ transfer functions were estimated correctly. First, MAP, $PETCO_2$, and CBFV can be measured in human recordings, but the true contribution of each input is not independently known. Second, when the inputs are related, a model may predict total CBFV well while assigning part of the response to the wrong input. Model fit therefore describes how well the output is reproduced, but it does not establish that the individual pathways were recovered correctly. Peng et al. also examined multivariable system identification for cerebral autoregulation (Multivariate System Identification for Cerebral Autoregulation), showing that multiple coherence was higher than the corresponding single-input coherence for MAP alone. Together, these studies provide a basis for two-input modeling, but they do not fully answer when a frequency-domain MISO model can recover the two pathways accurately.

A two-input MISO TFA model addresses this limitation by estimating the MAP-to-CBFV and $PETCO_2$-to-CBFV transfer functions together. The coherence measures used with this model answer different questions. Ordinary coherence describes the pairwise relationship between one input and CBFV without accounting for the other input. Partial coherence describes the remaining relationship between one input and CBFV after accounting for the second input. Multiple coherence describes how consistently MAP and $PETCO_2$ together are related to CBFV. Higher multiple coherence indicates that the two-input model explains the measured output

more closely, but it does not show by itself that the individual pathways were estimated accurately.

[Figure 1 placeholder: Comparison of SISO and MISO model structure]

A clear and consistent MISO TFA framework is therefore needed because preprocessing, spectral settings, the joint model solution, and diagnostic criteria can all affect the results. The framework must document these choices, return the MAP- and $PETCO_2$-related gain and phase estimates, and report the coherence and numerical diagnostics needed to interpret them. The computational details of this framework are described in the Methods.

Adding a second input does not automatically produce a more accurate model. MISO must estimate two pathway coefficients at each frequency from the same finite recording. If MAP and $PETCO_2$ become too similar within a frequency region, the model can have difficulty distinguishing their effects. In this situation, small spectral errors can produce large changes in the estimated transfer functions. Short recordings, weak $PETCO_2$ fluctuations, measurement noise, signal misalignment, and estimator settings may also affect this balance.

The two-input framework will first be applied to resting baseline recordings from NC adults. Comparing MISO with conventional SISO estimates will show how accounting for both measured inputs changes the estimated MAP- and $PETCO_2$-related CBFV dynamics. Partial and multiple coherence and the conditioning diagnostics will be used to evaluate whether the two-input estimates are supported by the available recordings. This application provides physiological relevance, but it cannot establish which model is closer to the underlying pathways because those pathways are not known in subject data.

Known-truth simulation provides the complementary evaluation. For each simulated family, prespecified MAP-to-output and $PETCO_2$-to-output transfer functions are defined first. Synthetic input time series are then generated, passed through these model pathways, and combined to create a synthetic output time series. MISO and SISO models are estimated using only the generated input and output observations, and their gain, phase, and complex transfer functions are compared directly with the prespecified transfer functions. Recording duration, noise, input relationships, timing alignment, and estimator settings are varied to determine how each condition affects pathway recovery. Comparing measurable properties of the NC recordings with these tested conditions will then show whether the physiological recordings resemble conditions that support stable two-input estimation.

The primary objective of this study is to develop and evaluate a reproducible two-input TFA framework that jointly estimates the MAP-to-CBFV and $PETCO_2$-to-CBFV dynamics. We will apply the framework to NC recordings and compare the resulting pathway estimates with conventional SISO analyses. We will then use known-truth simulations to evaluate pathway-

recovery accuracy across relevant recording and input conditions and determine whether the NC recordings occupy regions in which the two-input system is adequately identifiable. Together, these analyses will document the method, demonstrate its physiological application, and define the conditions that support reliable interpretation.